IX. Appendices

Appendix A - Classification Class Mapping and Split Manifest Summary

This appendix consolidates the class mapping and split identity that support the classification methodology and results. It records data-partition provenance and retraining status without reproducing the performance tables from Chapter VI.

Table A.1. SDI-4 four-class mapping and fixed seed-42 split

Serialized class	Train	Validation	Test	Total
Healthy	282	60	61	403
BG	139	30	29	198
WSSV	229	49	50	328
WSSV_BG	154	33	33	220
Total	**804**	**172**	**173**	**1,149**

The primary SDI-4 split was generated once with seed 42 and preserved for directly compared classification runs. Class order remained Healthy, BG, WSSV, and WSSV_BG across manifests, training class counts, output decoding, metric scripts, and mobile metadata. Historical candidate screening inspected test-partition Macro-F1, so the test partition was not completely independent of model-family selection.

Table A.2. Additional-source mapping and retraining status

Regime	Source mapping / construction	Retained test size	Evaluation status
EXT-3-Original	Tom_BT → Healthy; Den_Mang → BG; Dom_Trang → WSSV. No WSSV_BG category.	47	The mapped three-class model was retrained in this regime; this is not direct inference by the original four-class checkpoint.
SDI-4 + EXT-3-Original	Each source was split independently under the nominal 70/15/15 policy, then train+train, validation+validation, and test+test were merged. WSSV_BG is supplied only by SDI-4.	220	The four-class model was retrained on the integrated partitions. Source composition and class-prior changes remain explicit interpretation factors.

The integrated regime is therefore a partition-wise retraining experiment rather than zero-shot external inference. Legacy labels such as ShrimpDB-3 or Combined-4 are retained only as historical aliases; the primary dataset identifiers used throughout the thesis are EXT-3-Original and SDI-4 + EXT-3-Original.

Appendix B - Classification Experimental Configuration

This appendix records retained implementation constants and controls for the primary classification comparison. Values are reported only when preserved by the project evidence.

Table B.1. Primary classification training and checkpoint configuration

Item	**Recorded setting**	**Interpretation**
Reference architecture	YOLO26m-cls	Common backbone for the CE reference and final proposed configuration.
Input representation	Background-removed RGB images; 224 $\times$ 224	Offline foreground preparation was reused across directly compared classifiers.
Partition and seed	Fixed train/validation/test manifest; seed 42	Partition membership was not regenerated per directly compared run.
Optimizer and schedule	AdamW; initial LR 0.00125; cosine schedule; final LR factor 0.01	Shared selected-model training policy.
Training budget	Maximum 30 epochs; patience 15; batch 32; four workers; mixed precision enabled	Checkpoint selected from validation-side training records.
Training augmentation	RandAugment through retained Ultralytics classification configuration	Applied to training only; exact library-defined transforms remain configuration-bound.
Reference objective	Cross-entropy	Controlled reference condition.
Final configuration	ASL-LDAM + SimAM-DCFR	ASL-LDAM is training-only; feature recalibration remains in the exported network.
Candidate preference	Deployment-oriented preference approximately $\leq$15M parameters	Models above this preference may remain as higher-capacity diagnostic controls but are not automatically eligible as final mobile candidates.

Table B.2. ASL-LDAM and SimAM-DCFR implementation constants

Component	Recorded value	Role
SimAM energy constant	$\lambda = 10^{-4}$	Stabilizes the variance-dependent inverse-energy computation.
Texture path	Depthwise 3×3, groups $= C$, no bias; pointwise 1×1, no bias	Project-created local texture recalibration with preserved tensor shape.
Channel gate	GAP; 1×1 convolution, $C \to C$; sigmoid	Trainable channel-wise residual gate.
Fusion	$F' = F + \frac{1}{2} G \odot (F_{\text{SimAM}} + F_{\text{texture}})$	Residual integration of SimAM and project-created texture features.
ASL focusing	$\gamma_{pos} = 0,\ \gamma_{neg} = 4$	Leaves the true-class term unfocused and suppresses easy non-target terms.
LDAM	$m_{max} = 0.5$, scale $s = 30$	Applies a class-dependent margin to the true-class logit before softmax.
Insertion contract	$F' \in \mathbb{R}^{B \times C \times H \times W}$ before the original classification head	Preserves feature shape and four-logit output contract.

Only the SimAM path is parameter-free. The texture path and channel gate contain trainable convolutions. At inference, class selection uses the exported four-score output and deterministic class mapping; ASL-LDAM margins and loss weights are absent from the mobile graph.

Appendix C - Classification Candidate Inventory

This appendix preserves the screened model identities without reconstructing missing performance rows. The deployment-oriented preference is approximately 15M parameters or less; larger candidates remain useful diagnostic references when their evidence was retained.

Table C.1. TIMM candidate inventory

Candidate	Family	Retained evidence status	Deployment status
convnext_tiny_in22k	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobilenet_v3_large	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
efficientnet_b0	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
repvgg_a0	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate

Candidate	Family	Retained evidence status	Deployment status
efficientnet_v2_s	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
shufflenet_v2_x1_0	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
fastvit_t8	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
edgenext_xx_small	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobileone_s0	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobilevit_s	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobilenetv4_conv_small	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mnasnet_100	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
ghostnetv2_100	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
rexnet_100	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
squeezenet1_1	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
mobilenetv4_hybrid_medium	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate
efficientvit_m1	TIMM	Candidate identity retained; complete joint metric ledger not retained	Identity-only retained candidate

Table C.2. **Ultralytics classification candidate inventory**

Candidate	Family	Retained evidence status	Deployment interpretation
yolo26m-cls	YOLO26	Complete representative performance and efficiency row retained	Eligible under the final deployment-oriented preference
yolo26x-cls	YOLO26	Complete representative performance and efficiency row retained	Higher-capacity diagnostic control; above the approximately 15M preference
yolov8m-cls	YOLOv8	Complete representative performance and efficiency row retained	Above the approximately 15M preference in the retained candidate screen
yolov8s-cls	YOLOv8	Complete representative performance and efficiency row retained	Eligible by parameter preference
yolo11m-cls	YOLO11	Complete representative performance and efficiency row retained	Eligible by parameter preference
yolov8x-cls	YOLOv8	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolov8l-cls	YOLOv8	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo26l-cls	YOLO26	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo26n-cls	YOLO26	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo11x-cls	YOLO11	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo11s-cls	YOLO11	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo11n-cls	YOLO11	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolov8n-cls	YOLOv8	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo26s-cls	YOLO26	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed
yolo11l-cls	YOLO11	Candidate identity retained; complete joint row not retained	Identity-only; deployment eligibility not reconstructed

The inventory records screening provenance rather than a complete ranking. A missing metric row is not reconstructed from architecture name, parameter family, or later model performance.

Appendix D - Segmentation Annotation and Split Rules

This appendix records the final segmentation label semantics, the specimen-grouped subset, the expanded inventory, and the leakage boundary used to interpret the segmentation results.

Table D.1. Segmentation annotation semantics and validity boundaries

Element	Recorded rule	Validity boundary
Positive mask classes	BG and WSSV	WSSV_BG is an image-level classification category, not a third mask class.
Healthy images	No positive BG/WSSV polygon; retained in train, validation, and test as empty-label negatives	Healthy is not excluded from segmentation optimization; these images support negative-image false-positive evaluation.
Co-infection images	May contain BG masks, WSSV masks, or both	Multiple instances/classes can occur in one image.
Overlapping project polygons	When authored BG and WSSV polygons overlap, WSSV takes deterministic project-specific priority in overlapping pixels	This is an annotation convention for target construction, not a biological hierarchy and not a limitation of YOLO instance representation.
Annotation provenance	One annotator	No independent expert boundary review or inter-annotator agreement.
Specimen identity	Derived from compatible disease-folder and filename identifiers	Grouping can be enforced only where filenames expose a compatible specimen key.

Table D.2. Original filename-compatible grouped split statistics

Partition	Images	Specimens	Disease-labeled	Healthy	Masks	WSSV_BG images
Train	905	331	584	321	810	173
Validation	115	40	74	41	102	21
Test	129	45	88	41	119	26
Total	**1,149**	**416**	**746**	**403**	**1,031**	**220**

The 1,031 project-authored mask instances comprise 462 BG and 569 WSSV instances. The original 1,149-image subset is compatible with specimen grouping because its filename convention exposes 416 recoverable specimen groups.

Table D.3. Expanded instance-segmentation image inventory

Split	Images	Convention-matched images	Added unmatched-name images
Train	1,147	905	242
Validation	144	115	29
Test	161	129	32
Total	1,452	1,149	303

The 303 added images do not expose compatible specimen identity in their filenames. They were therefore allocated by image-level stratified random splitting before the partitions were merged. The expanded 1,452-image inventory must not be described as fully specimen-leakage-free: specimen-group separation is enforced for convention-matched images, while specimen identity cannot be enforced for the unmatched-name subset.

Group-overlap definition and leakage study

For a split strategy, train-test overlap is measured at specimen-group level as

$$\mathrm{GroupOverlap}(\%) = 100 \times \frac{|G_{\text{train}} \cap G_{\text{test}}|}{|G_{\text{test}}|}. \tag{D.1}$$

This quantity is not image overlap. The retained multi-seed comparison uses seeds 42, 123, and 3407 and reports mean $\pm$ standard deviation.

Table D.4. Leakage comparison across retained segmentation split policies

Split policy	Train-test overlap (%)	Full mAP50	Labeled mAP50	mAP50-95	Healthy FP (%)
Plain random image	98.7 ± 1.2	67.1 ± 1.3	67.7 ± 1.3	29.9 ± 2.4	5.4 ± 2.7
Stratified random image	94.0 ± 6.1	65.1 ± 6.0	65.9 ± 6.2	28.2 ± 1.6	9.8 ± 2.4
Stratified grouped specimen	0.0 ± 0.0	45.1 ± 2.7	47.3 ± 3.5	14.4 ± 2.1	23.6 ± 11.0

The higher random-split scores coincide with substantial specimen overlap and therefore cannot be interpreted as stronger leakage-safe generalization. The grouped split is the principal validity reference for the filename-compatible subset.

Appendix E - Attention Configuration Ledger

This appendix preserves both the protocol ledger and the retained parameter/timing evidence for segmentation attention configurations. The broader attention-family comparison is configuration-level evidence, not a strict one-factor ablation, because augmentation, training budget, early stopping, initialization, software versions, placement, and auxiliary objectives are not identical across all rows.

Table E.1. Recorded attention configuration protocols

Config.	Placement / objective	Aug. policy	Epochs (cfg./run)	Initialization	Versions
YOLO11n-seg reference	No added attention	Clean-light	100 / 60, early stop	yolo11n-seg.pt	8.4.61 / 2.10.0
CA→SimAM	P3/P4/P5	Strong	100 / 100	594/594 tensors	8.4.67 / 2.11.0
LKA→SimAM	P3/P4/P5	Clean-light	100 / 75, early stop	579/579 tensors	8.4.61 / 2.10.0
DPCA	P3/P4/P5	Strong	200 / 200	yolo11n-seg.pt	8.4.102 / 2.10.0
CoTEGate + BoundaryLite	P4 + auxiliary loss	Strong	200 / 184, early stop	yolo11n-seg.pt	8.4.95 / 2.10.0

Table E.2. Attention localization-efficiency ledger

Configuration	Parameters	Params (M)	Notebook ms/image	FPS
YOLO11n-seg reference	2,842,998	2.843	6.0	166.7
CA→SimAM	2,854,718	2.855	9.3	107.5
LKA→SimAM	2,963,510	2.964	8.6	116.3
DPCA	2,848,377	2.848	8.8	113.6
CoTEGate + BoundaryLite	2,736,562	2.737	8.2	122.0

These timing values are notebook/GPU diagnostics and are not Android latency. CA→SimAM adds only 11,720 parameters relative to the reference while providing the retained final attention refinement.

Table E.3. CA→SimAM placement study on YOLO11n-seg

Placement	Full mAP50	Lab. mAP50	Lab. mAP50-95	Recall	Params (M)	ms/image	FPS
P3/P4/P5	49.71	55.07	19.64	49.7	2.855	9.3	107.5
Neck concat	40.90	43.30	14.20	46.5	2.819	10.3	97.1
Head P3	40.90	43.40	14.60	41.5	2.692	9.9	101.0
Head P4	47.70	50.50	16.80	47.4	2.693	9.6	104.2
After C2PSA	43.80	45.80	16.90	45.6	2.697	10.3	97.1
Backbone P3	38.50	40.90	13.60	43.6	2.692	10.6	94.3
Neck P3/P4	44.30	47.30	15.90	43.6	2.721	10.4	96.2

P3/P4/P5 leads the retained localization and recall columns within this placement-controlled study. This placement result supports the selected multi-scale insertion pattern, while the broader attention-family table above remains non-identical in protocol.

Appendix F - Fourier Configuration and Spectrum Ledger

This appendix separates input-space W1 preprocessing from conventional augmentation and feature-domain Fourier refinement. It preserves the retained candidate screen, transform-order study, timing scope, and the direct paired spectrum analysis.

Table F.1. Retained Fourier candidate screen under the strict control

Configuration	Labeled mAP50	mAP50-95	Healthy FP (%)	Disease miss (%)	HScore (%)
No Fourier control	23.54	6.06	39.02	22.73	14.08
High-pass $\sigma = 50, \alpha = 0.05$	30.88	7.92	34.15	25.00	20.90
W1 high-pass $\sigma = 50, \alpha = 0.10$	34.51	8.85	63.41	19.32	21.08
High-frequency damping	32.11	10.68	39.02	52.27	17.40
Band-pass 12/60, $\alpha = 0.20$	29.52	8.39	26.83	43.18	17.21
Low-frequency flattening	27.87	8.52	14.63	46.59	16.66
Homomorphic filtering	26.12	9.16	41.46	23.86	15.06
Gaussian low-pass $\sigma = 25$	25.08	8.56	31.71	37.50	15.37

W1 uses a Gaussian low-pass mask with $\sigma = 50$ and residual enhancement $\alpha = 0.10$. It changes image intensities without geometric warping, so the associated polygon coordinates remain unchanged. Its effect is augmentation-regime dependent rather than uniformly beneficial.

Table F.2. Offline augmentation-Fourier order ledger

N=1 augmentation	Aug.-only mAP50	Higher Fourier order	Fourier mAP50	Δ (pp)
Translate 0.08	37.82	Augmentation $\rightarrow$ W1	35.00	-2.82
Scale 0.10	36.91	Augmentation $\rightarrow$ W1	31.85	-5.05
FlipLR 0.25	32.75	W1 $\rightarrow$ augmentation	36.51	+3.77
Rotation 3 degrees	31.08	W1 $\rightarrow$ augmentation	35.95	+4.87
Erasing 0.15	19.81	W1 $\rightarrow$ augmentation	29.07	+9.26
FlipUD 0.30	33.45	Augmentation $\rightarrow$ W1	34.51	+1.05

Table F.2 is an offline one-transform order probe. The separately recorded segmentation configuration field erasing=0.15 was inactive in the retained Ultralytics segmentation transform path; consequently, the offline erasing row must not be interpreted as evidence that random erasing executed inside the retained strong online augmentation policy.

Table F.3. Recorded Fourier timing scope

Method	Fourier preprocessing (ms/image)	YOLO prediction (ms/image)	Effective total (ms/image)
W1 high-pass	48.62	38.77	87.38
Erasing 0.10	0.00	12.69	12.69
FlipUD 0.30	0.00	12.69	12.69
Scale 0.10 + Erasing 0.10	0.00	12.82	12.82

Timing used the 129-image grouped test set, CUDA execution, 640-pixel inputs, ten warm-up images, and 119 timed images. These values are not Android runtime and are not interchangeable with classification screening timing.

Paired W1-augmentation spectrum protocol

Table F.4. Direct paired spectrum-study design

Protocol item	Retained setting	Interpretation boundary
Held-out images	24 images: six each from Healthy, BG, WSSV, and WSSV_BG	Balanced four-stratum diagnostic sample.
Matched augmentation draws	12 stochastic realizations per image	Same random realization is paired across processing paths.
Global paired observations	288 per global image metric	Supports within-image spectrum comparison, not an additional model benchmark.
Global metrics	Low/mid/high frequency energy share; radial spectral power; Laplacian variance; mean gradient magnitude	Measures redistribution of spatial/frequency content.
Lesion-region metrics	Lesion-region gradient where a disease mask exists	Healthy images have no lesion mask; lesion-region observation count is therefore lower.
Claim boundary	Seed-42 diagnostic mechanism analysis	Not causal proof and not multi-seed confirmation.

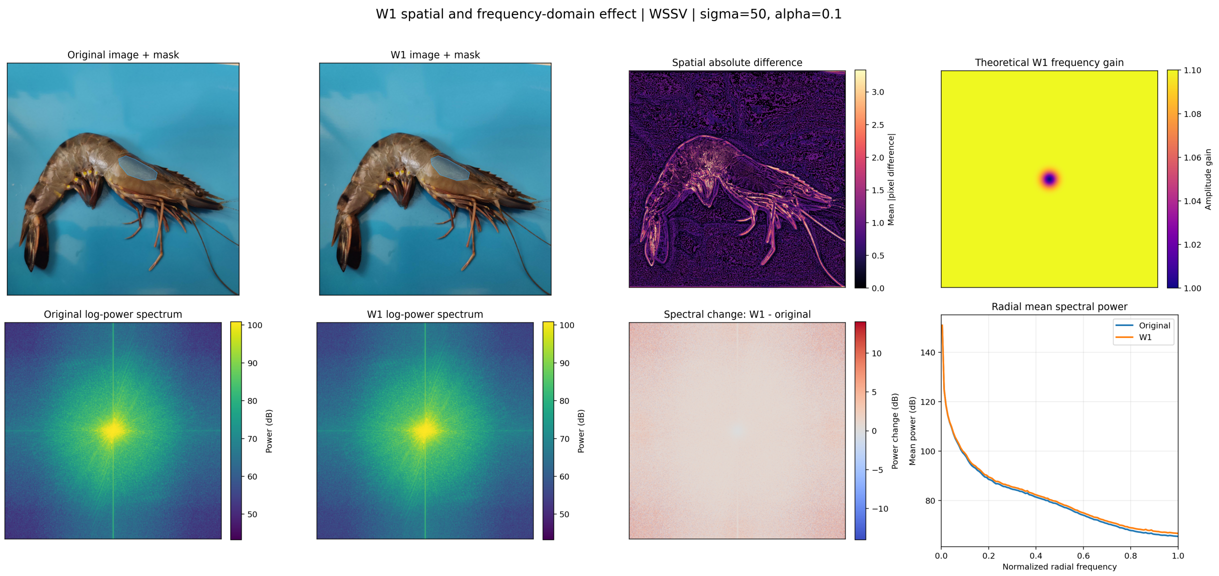


Figure F.1. High-resolution audit view of W1 spatial and frequency-domain behavior at $\sigma = 50$ and $\alpha = 0.10$. The panels show the original and W1 images, spatial difference, theoretical gain, log-power spectra, signed spectral change, and radial mean power.

Figure F.1 documents the preprocessing operator itself: W1 introduces a small residual high-frequency emphasis while preserving image geometry. The theoretical gain panel describes the linear filter before clipping, whereas the image and spectrum panels show the realized effect on a retained shrimp image.

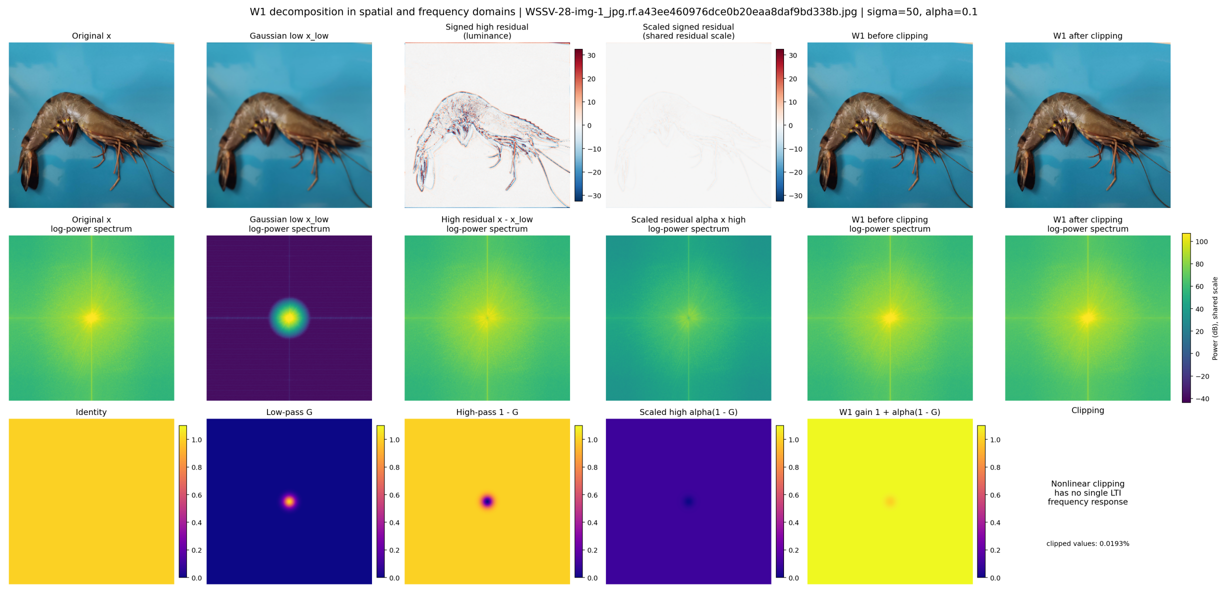


Figure F.2. Supplementary W1 decomposition showing the Gaussian low-pass component, signed high-frequency residual, scaled residual, pre-clipping output, post-clipping output, and corresponding frequency responses.

Figure F.2 separates the linear residual construction from nonlinear clipping. This distinction is important because clipping has no single linear time-invariant frequency response; the reported spectrum analysis therefore evaluates the actual processed images rather than assuming the theoretical gain fully determines the final spectrum.

Table F.5. Retained spectrum-effect summary under strong augmentation

Metric	Retained numerical evidence	Retained fraction	Interpretation
High-band energy share	W1 alone: +0.000670; incremental W1 after strong augmentation: +0.000151	22.5%	Most of W1's global high-band emphasis overlaps with or is redistributed by the strong pipeline.
Laplacian variance	Retained ratio reported from the paired study	21.1%	Edge-energy increment is strongly attenuated after strong augmentation.
Mean gradient magnitude	Retained ratio reported from the paired study	56.1%	A larger fraction of average gradient effect remains than for high-band share.
Lesion-region gradient	Retained ratio reported for disease-mask regions	approx. 79%	Local lesion-region edge emphasis is less attenuated than the global spectral summary.

The spectrum evidence supports partial redundancy of W1 under the retained strong augmentation policy. It does not establish a universal causal mechanism: the result is tied to the recorded seed-42 model, augmentation implementation, and paired diagnostic sample.

Feature-domain FDDEM remains a separate experimental direction; no final fully audited checkpoint-result pair was retained, so no numerical FDDEM outcome is added here.

Appendix G - Evaluation Metric Definitions

This appendix consolidates metric definitions used to interpret classification and segmentation results. Metrics from different tasks are not ranked against one another.

For N single-label classification samples, accuracy is

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i]. \tag{G.1}$$

For class k,

$$\text{Precision}_k = \frac{TP_k}{TP_k + FP_k}, \quad \text{Recall}_k = \frac{TP_k}{TP_k + FN_k}, \quad F1_k = \frac{2\,\text{Precision}_k\text{Recall}_k}{\text{Precision}_k + \text{Recall}_k}. \tag{G.2}$$

Macro-F1 assigns equal weight to the K classes:

$$\text{Macro-F1} = \frac{1}{K}\sum_{k=1}^{K} F1_k. \tag{G.3}$$

Cohen's Kappa is

$$\kappa = \frac{p_o - p_e}{1 - p_e}. \tag{G.4}$$

For segmentation,

$$\text{IoU} = \frac{|M_{pred} \cap M_{gt}|}{|M_{pred} \cup M_{gt}|}. \tag{G.5}$$

Average precision integrates the precision-recall curve for one class. mAP50 averages class AP at IoU 0.50; mAP50-95 averages AP over IoU thresholds 0.50 through 0.95 in increments of 0.05. Full-test evaluation includes all designated images, whereas labeled-only evaluation restricts the set to images containing at least one BG or WSSV mask.

Healthy false-positive rate and disease missed-image rate are

$$\text{FPR}_{healthy} = \frac{\#\{\text{Healthy images with at least one retained disease prediction}\}}{\#\{\text{Healthy images}\}}, \tag{G.6}$$

$$\text{MissRate}_{disease} = \frac{\#\{\text{diseased images with no retained prediction}\}}{\#\{\text{diseased images}\}}. \tag{G.7}$$

Mask-count mean absolute error is

$$\text{MAE}_{count} = \frac{1}{N}\sum_{i=1}^{N} |\hat{n}_i - n_i|. \tag{G.8}$$

The project-defined Healthy-aware score is

$$\text{HScore} = \text{mAP50}_{labeled} - 0.05\,\text{MAE}_{count} - 0.15\,\text{MissRate}_{disease} - 0.10\,\text{FPR}_{healthy}. \tag{G.9}$$

HScore is an internal decision criterion, not a standardized external benchmark. Percentage-form components must be represented on a consistent scale before combination.

For split-validity analysis, GroupOverlap is defined by (D.1) at specimen-group level. RTMDet recall values in the added comparator study were computed under the recorded RTMDet evaluation at IoU 0.50 and confidence 0.25, whereas YOLO mask recall is the Ultralytics-exported recall. These definitions are not treated as numerically identical. MMEngine evaluator timing is likewise not reported as pure RTMDet inference latency.

Appendix H - Corruption Protocol Inventory

This appendix keeps classification and segmentation corruption evidence separate. The tasks used different transform families, sample scopes, and reporting rules.

Table H.1. Classification corruption inventory

Family	Retained severity evidence	Reporting boundary
Impulse noise	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained; labels preserved
Gaussian noise	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained; not interchangeable with blur
Contrast reduction	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained
Defocus blur	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained
Low light	Clean plus severity levels 1-3 in qualitative panels	Aggregate Macro-F1 and Kappa retained

The exact transform parameters remain tied to archived transformation definitions rather than inferred from library defaults. The retained aggregate table does not provide a complete per-severity machine-readable ledger.

Table H.2. Segmentation corruption inventory and naming boundary

Family	Retained evidence	Interpretation
Color cast	Quantitative positive-gain row and qualitative examples	One of five selected largest positive gains
Turbidity	Quantitative positive-gain row and qualitative examples	Synthetic visibility and color attenuation
JPEG artifact	Quantitative positive-gain row and qualitative examples	Compression-specific stress condition
Overexposure	Quantitative positive-gain row and qualitative examples	Intensity clipping / illumination stress
Gaussian noise	Quantitative positive-gain row	Distinct from Gaussian blur
Gaussian blur	Qualitative panel only	No quantitative value may be substituted from the Gaussian-noise row

The segmentation table contains selected retained conditions rather than a complete corruption benchmark. It cannot establish improvement for unreported conditions, and its values are not directly comparable with classification corruption results.

Appendix I - Mobile Interface and Deployment Configuration

This appendix records the final conditional two-model Android deployment contract and the observational runtime protocol. The mobile evidence is operational feasibility evidence, not field disease-accuracy validation.

Table I.1. Final mobile model-bundle contract

Role	Deployed artifact / precision	Interface contract	Verified boundary
Classifier	yolo26m_asl_ldam_simam_dcfr_fp32.tflite; FP32	Input [1,224,224,3]; output [1,4]; order Healthy, BG, WSSV, WSSV_BG	Always executes first. Export size/hash are retained by the classification weight audit, which also flags an internal train-args naming conflict; export identity alone does not prove the final reported metrics.
Segmenter	yolo11n_seg_float16.tflite; FP16; retained size 5.65 MB	YOLO11n-seg + CA→SimAM at P3/P4/P5; 640 × 640 RGB; BG/WSSV instance masks	Invoked only for above-threshold BG/WSSV/WSSV_BG outputs. The current document set retains final artifact metadata and app usage, but no segmentation cryptographic hash is retained here.
Branch control	Application-side dispatch	Healthy or below threshold → stop; disease above threshold → segmentation	Predicted class controls execution and is not a verified field disease label.

The final classifier parameter count is not copied from the plain YOLO26m candidate row. The audited candidate export is 41,443,690 bytes with SHA-256 9834ecbaeee14878da2fd53a811005cfa2ed8192b94f9d8a7f5b732b19d1689d. The archived weight audit also preserves the train-args naming conflict, so this hash identifies the audited export but does not independently reproduce the final reported metrics.

Table I.2. Android device configuration ledger

Device	OS / processor as recorded	CPU	Memory as recorded	Storage	GPU
vivo Y100	Android 15; Snapdragon 685, reported up to 2.8 GHz	8 cores	16 GB reported configuration	128 GB	Not recorded
Samsung A32	Android 13; MediaTek Helio G80, 2.00 GHz	8 cores	6 + 6 GB reported configuration	128 GB	Not recorded
POCO M7 Pro 5G	Android 16; MediaTek Dimensity 7025 Ultra	8 cores	8 + 4 GB reported configuration	256 GB	Not recorded

Memory entries are preserved as reported/configured values. Values such as 6+6 GB and 8+4 GB are not rewritten as wholly physical RAM, and missing GPU identifiers are not guessed. All three devices used the same application build, the same model bundle, and the common configured four-thread CPU path. GPU delegate execution was disabled in the retained application configuration.

Table I.3. Runtime logging and branch semantics

Runtime item	Recorded behavior	Interpretation boundary
Classifier execution	Runs for every accepted image	Every usable runtime record includes classification.
No-seg branch	Healthy above threshold or below-threshold short circuit	No disease mask is generated.
Segmentation-triggered branch	Above-threshold BG, WSSV, or WSSV_BG dispatches the second model	Branch includes additional segmentation preprocessing/decoding/application work.
Timing fields	Application-recorded elapsed time, speed, FPS, and running averages	May include preprocessing, interpreter activity, postprocessing, mask decoding, and UI/application overhead; not pure NN latency unless independently scoped.
Ground-truth fields	ground_truth_label and is_correct left empty	No mobile field accuracy, sensitivity, specificity, prevalence, or clinical-validity claim.

Runtime item	Recorded behavior	Interpretation boundary
Sampling	21 application runs per member/device; manually uploaded real-world images	Image identity/order was not standardized across devices; device means are not a strict matched-image hardware ranking.
Platform	Android only	No iPhone/iOS test was performed because no compatible test device was available.

Table I.4. Observational branch-runtime summary

Device	All runs	No-seg branch	Seg-triggered runs	Seg-triggered mean	Overall mean
vivo Y100	21	14; 5.016 s mean	7	6.914 s	5.649 s
Samsung A32	21	2; 0.499 s mean	19	2.155 s	1.997 s
POCO M7 Pro 5G	21	13; 1.860 s mean	8	2.641 s	2.157 s

The Samsung A32 no-seg rows are two below-threshold short-circuit cases and are not a stable Healthy-branch hardware estimate. On vivo Y100, eight Healthy above-threshold runs averaged 5.024 s; on POCO M7 Pro 5G, ten Healthy above-threshold runs averaged 1.903 s. Overall device means depend strongly on branch composition and are therefore not used as a strict hardware ranking.

The retained operational photo set contains 21 usable real-world shrimp images collected across the three members' phone cameras, with variation in lighting, blur/softness, viewpoint, orientation, and background clutter. One unusably dark image was excluded. The set has no veterinary or laboratory ground truth and is an operational acquisition case study, not a field-accuracy benchmark.

Appendix J - Reproducibility and Artifact Checklist

This checklist records the current preservation status of research and deployment evidence. "Retained" means the evidence is present in the document/artifact set; it does not imply that every artifact has an independently verified cryptographic chain.

Table J.1. Re-audited reproducibility and artifact checklist

Artifact / control	Status	Evidence boundary or required action
Primary class mapping and counts	Retained	SDI-4 four-class order and fixed split counts are documented consistently.

Artifact / control	Status	Evidence boundary or required action
Fixed seed and split policy	Retained	Seed 42 and primary classification partition statistics are retained.
Classification training configuration	Retained	Optimizer, schedule, budget, augmentation family, and checkpoint logic are recorded.
Representative candidate ledger	Partially retained	Complete joint rows exist for five Ultralytics candidates; remaining TIMM/YOLO joint metrics are not reconstructed.
Primary classifier checkpoint / export identity	Partially retained	Checkpoint and export hashes exist, but the archived weight audit records a train-args naming conflict; independent architecture/evaluation verification remains required before using the candidate weight as sole proof of the final 91.01% Macro-F1 result.
Classifier FP32 TFLite metadata	Retained	Filename, input/output shapes, class order, byte size, and SHA-256 are preserved by the weight audit; the explicit hash is reproduced with the deployment contract in Appendix I.
Primary prediction-level classification file	Not retained	Required to regenerate classwise reports, paired errors, and confusion matrices from frozen predictions.
Primary classwise report	Not retained	Must be regenerated only from frozen prediction-level evidence.
Primary count and normalized confusion matrices	Not retained	Must not be reconstructed from aggregate metrics.
Classification corruption aggregates	Retained	Five aggregate corruption families are documented.
Complete per-severity corruption ledger	Partially retained	Qualitative severity panels exist, but complete machine-readable per-severity metrics were not retained.
Original segmentation annotation statistics	Retained	1,149 images, 416 specimen groups, 746 disease-labeled images, 403 Healthy negatives, and 1,031 masks are documented.
Expanded segmentation inventory	Retained	1,452-image inventory and 303 unmatched-name additions are documented with the mixed grouped/stratified validity boundary.
Independent expert mask review	Not retained	Multi-annotator review and inter-annotator agreement remain recommended.

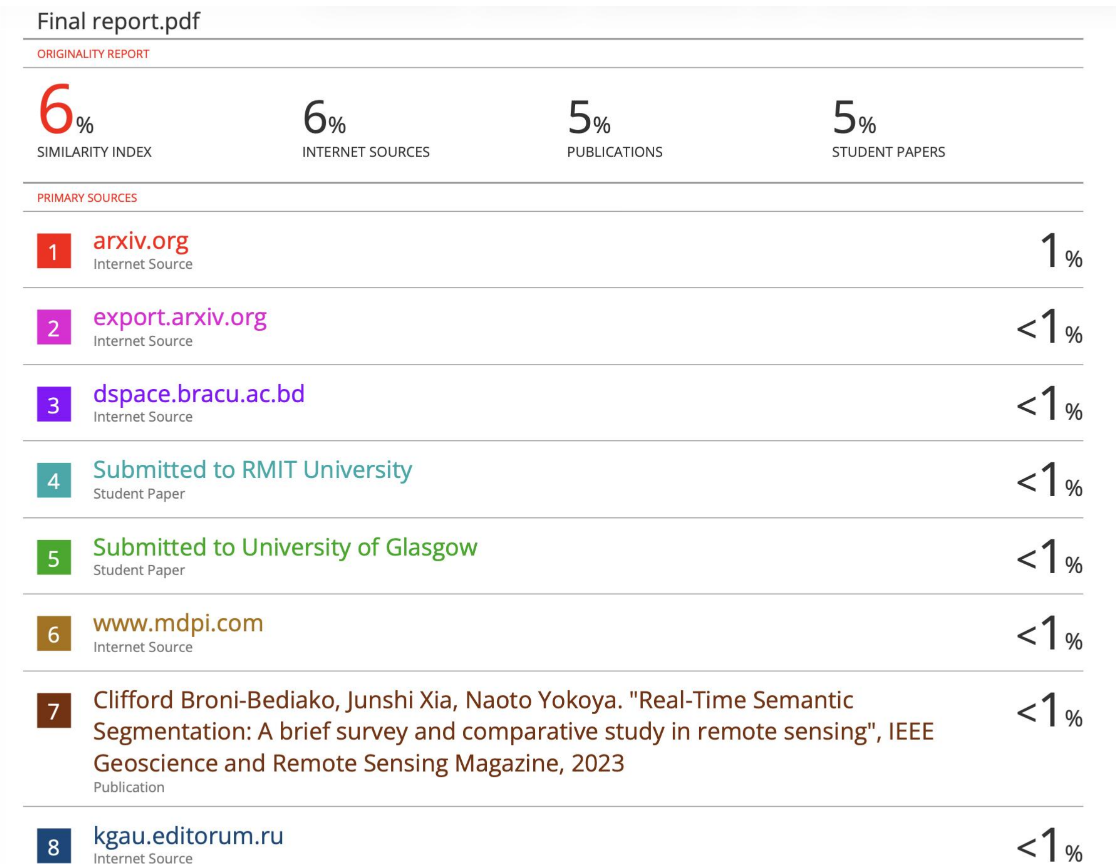
Artifact / control	Status	Evidence boundary or required action
Attention protocol ledger	Retained	Placement, augmentation, epochs, initialization, and software versions are documented.
Attention parameter/timing ledger	Retained	Reference and four attention configurations retain parameter counts and notebook timing; values are not Android latency.
CA→SimAM placement study	Retained	Seven insertion patterns are documented; P3/P4/P5 is the retained final placement.
Strict matched broad attention ablation	Not retained	Broader attention-family rows use non-identical protocols; future causal comparison should equalize all non-attention controls.
Fourier candidate and timing ledgers	Retained	W1 parameters, candidate metrics, transform order, and CUDA timing scope are documented.
Paired W1 spectrum study	Retained	24 images × 12 matched realizations = 288 paired global observations, with retained high-band and gradient-effect summaries.
Final audited FDDEM checkpoint/result	Not retained	No numerical FDDEM claim should be made without a frozen checkpoint-result pair.
RTMDet-Ins-S validation/test evidence	Retained	Validation trajectory, epoch-86 checkpoint selection, and test comparator row are retained; cross-framework recall definitions remain non-identical.
Final segmentation mobile metadata	Partially retained	YOLO11n-seg + CA→SimAM, FP16 filename, 5.65 MB size, decoder contract, and app use are documented; segmentation artifact hash is not retained here.
Three-device Android runtime logs	Retained	Three CSV logs with 21 runs each and the common application/model configuration are available; branch mixes differ.
Three-camera operational image set	Retained	21 usable photographs are retained as operational inputs; no veterinary/lab ground truth is available.
Multi-seed final result distribution	Not retained	Run three to five immutable-seed repetitions for uncertainty estimation.
Environment lockfile and dependency hash	Partially retained	Key framework versions are recorded for several runs; a unified lockfile and environment hash remain recommended.

Artifact / control	Status	Evidence boundary or required action
Automated metric and mobile regression suite	Recommended	Should verify class order, preprocessing parity, tensor shapes, known-image outputs, branch routing, and mask decoding.
Calibration and confidence communication report	Recommended	Needed before treating output confidence as a calibrated user-facing probability.

The checklist intentionally distinguishes artifact identity from result reproduction. In particular, a real cryptographic hash is recorded for the audited classifier export, while no segmentation hash is invented.

Appendix K - Supplementary Submission Evidence

This appendix preserves two administrative evidence artifacts associated with the capstone submission. They document the originality-check summary and the conference notification for the classification paper. These records support submission traceability but do not replace experimental evidence, peer-reviewed sources, or technical validation.


Final report.pdf

ORIGINALITY REPORT

6%	6%	5%	5%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	arxiv.org Internet Source	1%
2	export.arxiv.org Internet Source	<1%
3	dspace.bracu.ac.bd Internet Source	<1%
4	Submitted to RMIT University Student Paper	<1%
5	Submitted to University of Glasgow Student Paper	<1%
6	www.mdpi.com Internet Source	<1%
7	Clifford Broni-Bediako, Junshi Xia, Naoto Yokoya. "Real-Time Semantic Segmentation: A brief survey and comparative study in remote sensing", IEEE Geoscience and Remote Sensing Magazine, 2023 Publication	<1%
8	kgau.editorum.ru Internet Source	<1%

Figure K.1. Turnitin originality report summary for the capstone report. The captured report shows a 6% similarity index; this screenshot is retained as submission evidence and is not interpreted as a technical performance result.

Appendix K - Supplementary Submission Evidence (continued)

TỔ CHỨC GIÁO DỤC FPT

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Fw: ISBM 2026 notification for paper 301

Vinh Nguyen Dinh <VinhND29@fe.edu.vn> lúc 14:58 28 tháng 7, 2026
Đến: Nhan Huu Tran <nhanthce181655@fpt.edu.vn>

From: isbm2026@easychair.org <isbm2026@easychair.org> on behalf of ISBM 2026 <isbm2026@easychair.org>
Sent: 28 July 2026 13:19
To: Vinh Nguyen Dinh <VinhND29@fe.edu.vn>
Subject: ISBM 2026 notification for paper 301

LƯU Ý: Đây là email gửi từ bên ngoài Công ty. Anh/chị không bấm vào link hoặc file đính kèm nếu không biết người gửi cũng như nội dung đính kèm.
CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dear Vinh Nguyen Dinh,

Paper ID: 301

Title: ASL-LDAM with SimAM-DCFR Attention for Robust YOLO-Based Shrimp Disease Image Classification Under Noisy Imaging Conditions

Greetings .. !!

Congratulations! On behalf of the Program Committee of ISBM 2026 – Bangkok, Thailand, we are pleased to inform you that your paper has been accepted for oral presentation at the conference and for publication in the Springer LNNS (Lecture Notes in Networks and Systems) series, subject to the fulfillment of all required guidelines set by Springer. Please note that an accepted paper will be published in the Springer proceedings only if the final (camera-ready) version is submitted along with the payment details, including the transaction reference number, and meets the quality standards prescribed by Springer. We extend our warm congratulations and look forward to your participation in ISBM 2026 – Bangkok, Thailand.

Please follow the instructions in different sections for registration and participation -

Section A - Preparation of CRC (Camera-Ready Copy) – Final Manuscript for Springer (LNNS Series)

As an author you are requested to strictly adhere to the following points during the preparation of the final manuscript (CRC) to avoid any last-minute revert backs or objections from the publisher (Springer).

1. Format your paper using the official Springer template available at https://www.isbmconf.com/call-for-papers#downloads, and submit the final camera-ready copy (CRC) as a PDF without page numbers, headers, or footers.

2. In the author section, include only names, college/organization, city, and emails; do not mention designations, photos, or biographies.

3. Incorporate all reviewer comments into the final CRC, as no further review will be conducted after submission.

4. Ensure the manuscript is original, accurate, and free from plagiarism; rewrite any flagged content, and take full responsibility for the paper's correctness and integrity.

Figure K.2. ISBM 2026 notification for Paper 301, "ASL-LDAM with SimAM-DCFR Attention for Robust YOLO-Based Shrimp Disease Image Classification Under Noisy Imaging Conditions." The notification records acceptance for oral presentation and conditional publication, subject to the stated camera-ready, registration, payment, and Springer quality requirements.

Appendix L - Public Classification Repository

The public repository for the CVio shrimp-disease classification branch is available at: https://github.com/hntnhan1111-ayai/CVio_Shrimp_Disease_Classification_Capstone_SU26

The repository is publicly reachable and preserves project-level documentation and provenance requirements. Its current main branch states that integrated metrics and runtime code remain subject to verification gates; repository availability therefore does not expand the evidential scope of the thesis. Numerical claims remain bounded by the dataset identifiers, split manifests, seeds, configurations, checkpoints, software versions, and retained result files documented in the thesis and its evidence bundle.

The instance-segmentation attention and Fourier branches are reported separately and must not be inferred to be fully represented by this classification repository.